

Familiar Faces

Integrative Industry Synthesis

A Human-Supervised AI Prototype for Community Paramedicine Outreach Planning

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EDUCATIONAL SYNTHETIC-DATA PROTOTYPE

Not approved for clinical use, operational deployment, eligibility decisions, or autonomous patient contact.

Executive Summary

Familiar Faces is a human-supervised artificial intelligence prototype that explores how a community paramedicine program might prioritize outreach review for people who frequently use emergency medical services. Frequent emergency use may reflect unresolved barriers involving primary care, transportation, medication access, housing, behavioral health, or other social needs. Research on frequent emergency department users suggests that coordinated interventions can reduce utilization for some populations, but findings vary and do not justify assuming that every frequent user needs the same intervention (Althaus et al., 2011). The prototype therefore frames prediction as a limited screening aid for authorized staff, not as a clinical judgment or eligibility decision.

The system integrates descriptive analysis, supervised machine learning, fairness auditing, controlled nonclinical drafting, automated safety checks, and mandatory human review. It operates on 3,000 entirely synthetic EHR-like records and contains no real protected health information. Logistic regression was selected over random forest because it slightly outperformed the alternative while remaining easier to explain. Its test ROC AUC was 0.685 and average precision was 0.579, which represent a moderate advisory signal rather than deployment-quality evidence. Seven subgroup comparisons exceeded the project's monitoring threshold, producing a governance status of **REVIEW_REQUIRED**. These results demonstrate successful technical integration while also showing why model outputs must remain subordinate to accountable human judgment.

1. Industry Problem and Scope

The industry context is emergency medical services and community paramedicine. Traditional emergency response is designed to stabilize immediate problems, but repeated calls may originate from needs that cannot be resolved during a single encounter. Community paramedicine programs are positioned to coordinate nonemergency follow-up, connect people with existing resources, and complement rather than replace primary or emergency care. The realistic AI opportunity is not to diagnose patients. It is to help authorized personnel organize a review queue and prepare consistent planning material so that limited outreach capacity can be used more deliberately.

The system's goal is consequently narrow: estimate simulated outreach response, surface records for human review, and generate structured nonclinical drafts. It does not predict medical need, future deterioration, program eligibility, or expected clinical benefit. It cannot contact a patient, approve services, or recommend diagnosis or treatment. Those boundaries make the design appropriate to an educational capstone and reduce the risk that a technically successful prototype will be mistaken for an operational clinical tool.

2. Integration of Prior Capstone Projects

Project 4: machine-learning workflow. The NEMSIS cardiac-arrest project established the analytical foundation for this synthesis. It required careful cohort definition, documented preprocessing, reproducible train-test separation, comparison of models, and interpretation of evaluation metrics in an EMS context. Familiar Faces carries those practices forward through fixed random seeds, a stratified split, identical evaluation partitions, and explicit comparison of logistic regression with random forest. Project 4 also reinforced that predictive association is not causation. Accordingly, feature importance in this project is presented as a model explanation and not as proof that any barrier causes outreach engagement.

Project 5: deep-learning systems. The Seattle Fire 911 forecasting project demonstrated the value of recurrent neural networks and LSTMs when the problem contains sufficient sequential structure. It also taught an equally important design lesson: advanced architecture should follow the problem rather than precede it. Familiar Faces uses a small tabular synthetic dataset and requires transparent review of individual factors. A deep neural network would add complexity without a persuasive data or performance justification. Choosing simpler classifiers is therefore an intentional transfer of learning from Project 5, not an omission of cross-domain knowledge.

Project 6: generative AI. The NEISS narrative transformer project showed how generated text can vary, reproduce undesirable patterns, or appear more authoritative than its evidence supports. That experience shaped the decision to use deterministic, template-constrained drafting in this prototype instead of a live language model. Drafts can summarize documented barriers and matched resources, but they cannot invent facts or produce clinical language. Automated validation checks each draft for prohibited content, demographic references, and the required human-review statement. The generative-AI contribution is thus expressed through safe content design and validation, while the known risks of unbounded generation remain outside the system boundary.

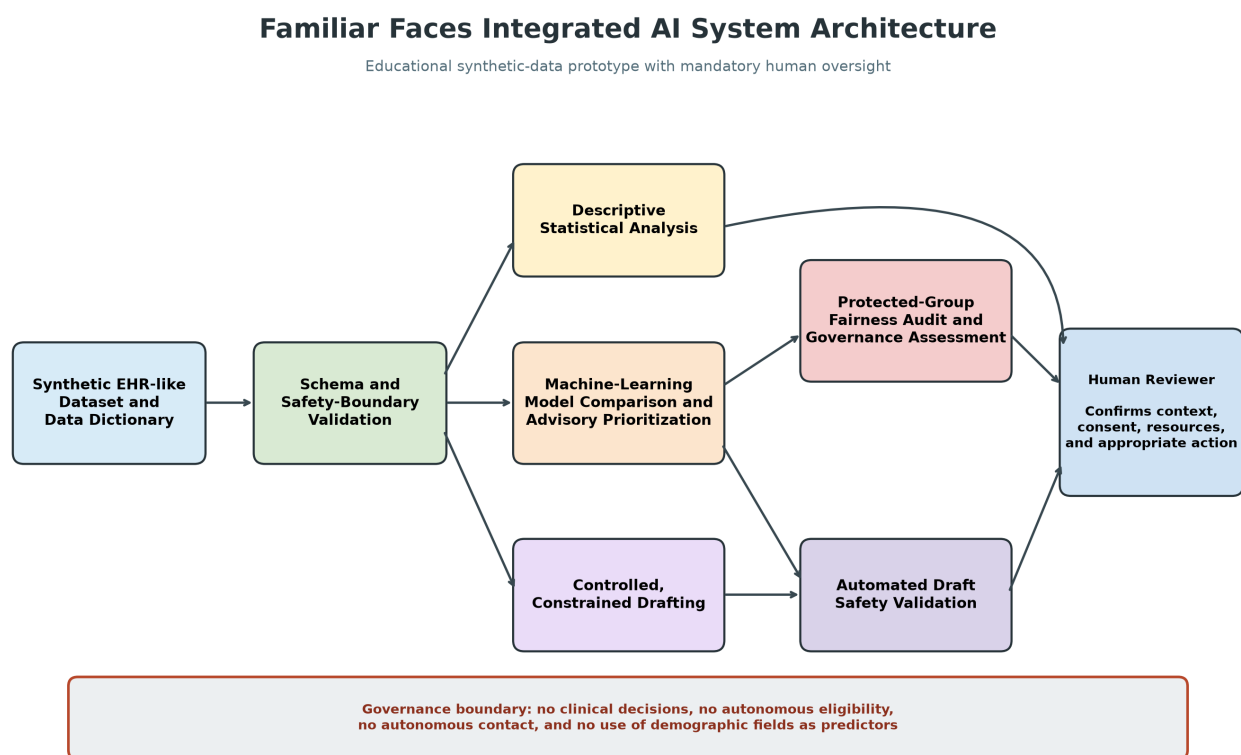
Project 7: agentic AI. The earlier Familiar Faces outreach agent supplied the operational pattern for tool use, permission checking, explicit state, and human escalation. That project demonstrated that an agent should stop when permission is denied or uncertain, urgent symptoms are present, a patient cannot be found, or a tool fails. Project 8 translates those lessons into a broader system architecture: analytical outputs feed controlled case materials, safety rules constrain downstream actions, and the final action boundary remains with authorized personnel. Integrating these projects produced one coherent workflow rather than four disconnected technical demonstrations.

3. System Architecture

The architecture separates data preparation, analysis, model evaluation, fairness governance, drafting, safety validation, and human review. Synthetic records first pass schema and quality checks. Descriptive analysis then provides context for model development. Two classifiers are trained on the same stratified partition, and the selected model produces probabilities used only to order a review queue. Protected demographic fields are excluded from model training and generated material; they are retained in a segregated audit path for post-model disparity assessment. Controlled drafts cannot proceed to any real-world action, because human approval is a required terminal boundary.

Figure 1

Familiar Faces Human-Supervised System Architecture



Note. Protected attributes follow an audit-only path. No component authorizes autonomous outreach or clinical action.

4. Data, Modeling, and Design Rationale

The dataset contains 3,000 synthetic records and 22 documented fields. Sixteen nonprotected fields are available to the classifiers. The analysis uses fixed seeds and a 75/25 stratified train-test split so that results are reproducible and the positive-class proportion is represented in both partitions. Logistic regression and random forest were evaluated with ROC AUC, average precision, accuracy, precision, recall, and F1. Multiple metrics are necessary because a single aggregate score can conceal operationally important tradeoffs between missed opportunities and unnecessary staff review.

Logistic regression was selected because it achieved slightly stronger test performance than random forest and offers coefficients that can be inspected more directly. The choice favors interpretability, stability, and governance over marginal complexity. Its predicted probabilities are not converted into automatic action. They support prioritization within a human workflow where staff can consider context unavailable to the model. The outcome itself is simulated outreach response, which is deliberately less consequential than a clinical target but still insufficient for deployment because synthetic labels cannot establish real-world validity.

5. Results and Evaluation

Table 1

Integrated Prototype Results

System metric	Result
Synthetic records	3,000
Selected model	Logistic regression
Test ROC AUC	0.685
Average precision	0.579
Test F1	0.558
Fairness governance status	REVIEW_REQUIRED
Comparisons requiring review	7
Small audit groups excluded	2
Drafts passing safety validation	12 of 12
Human review required	Yes

The pipeline executed successfully and produced all expected artifacts. Logistic regression achieved a test ROC AUC of 0.6852, average precision of 0.5791, accuracy of 0.6467, and F1 of 0.5576. Random forest achieved a ROC AUC of 0.6783 and F1 of 0.5179. The selected model therefore offered the stronger overall balance, but neither result supports autonomous prioritization. The controlled drafting stage produced 12 case materials, and all 12 passed automated safety validation. This confirms that the implementation behaves as designed; it does not demonstrate clinical effectiveness.

6. Ethics, Fairness, and Accountability

The central ethical concern is that an outreach model could distribute attention unequally or encode existing differences in access to care. Excluding protected attributes from prediction is not enough to guarantee fairness because other variables can act as proxies and the outcome label can reproduce structural inequity. Obermeyer et al. (2019) demonstrated that a health-management algorithm generated substantial racial bias even though its proxy outcome appeared predictive. Familiar Faces therefore separates prediction from subgroup auditing and treats observed disparities as reasons for investigation, not as proof that the system is fair.

The audit compares review-flag rates, true-positive rates, and false-positive rates across eligible groups. Groups with fewer than 25 test records are excluded from disparity calculations to avoid overinterpreting unstable estimates, but exclusion is documented rather than hidden. Seven comparisons exceeded the project threshold, so the system returns **REVIEW_REQUIRED**. That status is a governance signal, not a legal conclusion or a score that can be averaged away.

Responsible deployment also requires accountability beyond the model. NIST's AI Risk Management Framework emphasizes governance, contextual mapping, measurement, and ongoing management across the AI lifecycle (Tabassi, 2023). The prototype reflects that reasoning through documented scope, reproducible evaluation, audit outputs, human review, prohibited-use statements, and explicit escalation. Core safeguards include synthetic data only, no autonomous contact, no diagnosis or treatment recommendations, no eligibility decisions, and no use of demographic fields in prediction or drafting.

7. Tradeoffs, Constraints, and Deployment Requirements

Several tradeoffs shaped the design. Logistic regression sacrifices some ability to represent nonlinear relationships in exchange for clearer explanations and simpler validation. Deterministic drafting sacrifices the flexibility of a live language model in exchange for repeatability and tighter content control. Audit-only demographic data support disparity detection but also create privacy and governance obligations. Human review limits scale and introduces human variation, yet it preserves accountability where errors could affect access to support. These choices are appropriate for a high-consequence domain in which efficiency cannot be the only objective.

The largest constraint is external validity. Synthetic records can demonstrate code paths, metrics, and governance logic, but they cannot establish prevalence, causal relationships, model calibration, subgroup performance, or program benefit in a real population. Moderate model performance, small audit groups, and the **REVIEW_REQUIRED** result further limit interpretation. The project has not undergone clinical, legal, privacy, cybersecurity, accessibility, usability, or deployment validation.

Before any pilot, an organization would need authorized data-governance procedures; privacy and legal review; security controls and audit logging; independent model validation; calibration and subgroup analysis on representative data; stakeholder and community participation; documented reviewer training; appeal and escalation procedures; continuous performance and fairness monitoring; and assigned responsibility for incidents and model retirement. Outreach effectiveness would also need prospective evaluation against outcomes that matter to patients and programs, not merely response probability.

8. Professional Reflection and Conclusion

This synthesis changed my understanding of AI development from building an accurate component to designing an accountable system. My EMS experience made the problem recognizable, but the sequence of capstone projects supplied the technical discipline to address it responsibly: Project 4 contributed reproducible analytics, Project 5 clarified when not to use deep learning, Project 6 exposed the risks of unconstrained generation, and Project 7 supplied the permission and escalation pattern. The resulting artifact demonstrates cross-domain fluency while remaining honest about what the evidence does not support.

Familiar Faces is valuable as a portfolio piece because the code, executed notebook, architecture, model comparison, fairness artifacts, controlled drafts, and report all tell the same story. The system works as an

educational prototype, and its limitations are part of the result rather than footnotes to it. The appropriate next stage is stakeholder and governance evaluation, followed by carefully authorized research if the concept remains justified. It must not be used to make decisions about real people or to authorize clinical, eligibility, or outreach actions.

References

- Althaus, F., Paroz, S., Hugli, O., Ghali, W. A., Daeppen, J. B., Peytremann-Bridevaux, I., & Bodenmann, P. (2011). Effectiveness of interventions targeting frequent users of emergency departments: A systematic review. *Annals of Emergency Medicine*, 58(1), 41-52.e42. <https://doi.org/10.1016/j.annemergmed.2011.03.007>
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447-453. <https://doi.org/10.1126/science.aax2342>
- Tabassi, E. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.AI.100-1>